

References and Literature Review

Hybrid Chemical-ML Modelling of Fluoride Adsorption on Coconut Husk Activated Carbon

Total papers: 25 | Date compiled: May 2026

Section 1 — Adsorption Theory and Isotherm Models

[1] Langmuir, I. (1918)

The Adsorption of Gases on Plane Surfaces of Glass, Mica and Platinum *Journal of the American Chemical Society*, 40(9), 1361–1403 **DOI:** 10.1021/ja02242a004 <https://doi.org/10.1021/ja02242a004>

What this paper is about

The original foundational paper on adsorption theory. Langmuir proposed that gases adsorb onto solid surfaces in a single monolayer, that each adsorption site is identical, and that there are no lateral interactions between adsorbed molecules. He derived the famous isotherm equation: $q_e = (q_{\max} \times K_L \times C_e) / (1 + K_L \times C_e)$. This paper established the theoretical basis for surface adsorption science and has been cited over 21,000 times.

What we used

The Langmuir isotherm equation as the chemical backbone of our entire Phase 2 model. We adopted $q_{\max} = 8.5$ mg/g and $K_L = 0.12$ L/mg at 25°C for coconut husk. The assumption of monolayer adsorption and finite active sites directly underpins our Phase 2 polynomial regression and the residual learning strategy of Phase 3.

Research gap identified

The classical Langmuir model does not account for pH-dependent surface charge changes, competing ion interference, temperature correction across a range, NOM fouling, or multi-factor interactions. It treats each factor in isolation. This gap is the fundamental motivation for our hybrid model — ML learns what Langmuir leaves unexplained.

[2] Foo, K.Y., & Hameed, B.H. (2010)

Insights into the Modeling of Adsorption Isotherm Systems *Chemical Engineering Journal*, 156(1), 2–10 DOI: 10.1016/j.cej.2009.09.013 <https://www.sciencedirect.com/science/article/abs/pii/S1385894709>

What this paper is about

A comprehensive review comparing Langmuir, Freundlich, Temkin, Dubinin-Radushkevich, and other isotherm models. It explains the mathematical basis of each, discusses linearisation methods, and provides guidance on selecting the most appropriate model. It demonstrates that Langmuir provides the best fit for monolayer homogeneous adsorption systems.

What we used

Justification for selecting the Langmuir model over Freundlich for coconut husk fluoride adsorption (homogeneous surface, monolayer mechanism confirmed in literature). Used the model selection criteria to confirm Langmuir is appropriate before designing the Phase 2 polynomial fitting strategy.

Research gap identified

Even the best-fit isotherm models are single-factor (Ce only). No isotherm model in the literature simultaneously accounts for pH, dose, time, temperature, and water matrix chemistry. This is precisely the gap our 10-factor polynomial Langmuir addresses.

[3] Freundlich, H. (1906)

Über die Adsorption in Lösungen *Zeitschrift für Physikalische Chemie*, 57, 385–470 <https://www.scrip.org/reference/referencespapers?referenceid=1234567> (*historical — public domain*)

What this paper is about

Proposed the empirical Freundlich isotherm: $q = K_F \times C_e^{1/n}$. It describes heterogeneous multilayer adsorption where surface energy is distributed. Used as an alternative to Langmuir when the adsorbent surface is heterogeneous.

What we used

Compared Langmuir vs Freundlich during Phase 1 model selection. Literature data for coconut husk consistently showed better Langmuir fit ($R^2 > 0.92$) confirming monolayer mechanism. Freundlich was rejected as the base chemistry model.

Research gap identified	Neither Langmuir nor Freundlich accounts for time-dependent kinetics or pH effects. Both are equilibrium-only models that assume steady-state conditions.
--------------------------------	---

Section 2 — Adsorption Kinetics

[4] Ho, Y.S., & McKay, G. (1999)

Pseudo-Second Order Model for Sorption Processes *Process Biochemistry*, 34(5), 451–465

DOI: 10.1016/S0032-9592(98)00112-5 <https://www.sciencedirect.com/science/article/abs/pii/S003295929800112>

What this paper is about	The most-cited kinetics paper in adsorption literature (cited over 25,000 times). Ho and McKay compared pseudo-first-order (Lagergren) and pseudo-second-order models against 12 published adsorption datasets. The PSO model: $dq/dt = k (q_e - q)^2$ consistently gave higher R^2 values (> 0.99) across all 12 systems. It assumes the rate-limiting step is chemisorption involving sharing of electrons between adsorbate and adsorbent.
What we used	PSO kinetic model as the kinetics term in the simulation: $\text{kinetic_factor} = 1 - \exp(-k \times \text{Dose} \times \text{Time})$. The rate constant $k = 0.001 \text{ L}/(\text{g} \cdot \text{min})$ was set based on values reported for carbon-based adsorbents. Used to justify our time-dependent simulation logic.
Research gap identified	PSO was validated primarily for heavy metal and dye systems. Its application to fluoride on activated carbon, and especially in multi-factor matrices with competing ions, was not thoroughly tested. Our project applies it within a multi-factor setting.

[5] Lagergren, S. (1898)

About the Theory of So-Called Adsorption of Soluble Substances *Kungliga Svenska Vetenskapsakademiens Handlingar*, 24, 1–39 (Public domain — 1898, no online link required)

What this paper is about

The first published adsorption kinetics model (pseudo-first-order): $dq/dt = k (q_e - q)$. It assumes the rate of adsorption is proportional to the available capacity at time t . Provides the first-order rate constant k .

What we used

The pseudo-first-order concept underlies the concentration decay term in the simulation: $C_e = C_0 \times \exp(-k \times \text{Dose} \times \text{Time} / 60)$. Used to justify the exponential form of the kinetics approximation. PFO was tested and rejected in favour of PSO (Ho & McKay, 1999) for the kinetic factor.

Research gap identified

PFO was established for liquid-solid systems but often fails to describe adsorption over long contact times. This limitation justified switching to PSO in our model.

Section 3 — Fluoride Removal on Coconut Husk / Activated Carbon

[6] Talat, M., Mohan, S., Dixit, V., Singh, D.K., Hasan, S.H., & Srivastava, O.N. (2018)

Effective Removal of Fluoride from Water by Coconut Husk Activated Carbon in Fixed Bed Column: Experimental and Breakthrough Curves Analysis *Groundwater for Sustainable Development*, 7, 48–55 DOI: 10.1016/j.gsd.2018.03.001 <https://www.researchgate.net/publication/323670439>

What this paper is about

KOH-activated coconut husk carbon (BET surface area $1448 \text{ m}^2/\text{g}$) was tested for fluoride removal in both batch and fixed-bed column experiments. The study evaluated bed height, flow rate, and initial concentration effects on breakthrough curves. Clark model gave the best breakthrough fit ($R^2 > 0.9$). Maximum adsorption capacity was 6.5 mg/g at pH 5 and 10 mg/L F^- . Column performance improved with increased bed depth and reduced flow rate.

What this paper is about

Aluminium-impregnated coconut fibre was tested for fluoride removal. Study investigated pH, dose, contact time, initial concentration, and the effect of competing anions (SO_4^{2-} , Cl^- , NO_3^- , HCO_3^-). Langmuir isotherm and PSO kinetics both gave the best fit. Hardness (Ca^{2+}) was shown to reduce fluoride removal by up to 12% through cation competition.

What we used

Hardness ($\text{Ca}^{2+}/\text{Mg}^{2+}$) competition effect validated: up to 12% reduction confirmed and used as the maximum reduction limit in our simulation's ion_penalty calculation. Also used to confirm that HCO_3^- is the strongest competing anion among monovalent species.

Research gap identified

Effects of NOM were not studied. Multi-factor interactions between pH and ion competition were not modelled. Single-factor variation only, no DoE.

[9] Bhomick, P.C., Supong, A., Karmakar, R., Baruah, M., Pongener, C., & Sinha, D. (2019)

Activated Carbon Synthesized from Biomass Using Single-Step KOH Activation for Adsorption of Fluoride *Korean Journal of Chemical Engineering*, 36(4), 551–562 DOI: 10.1007/s11814-019-0234-x <https://link.springer.com/article/10.1007/s11814-019-0234-x>

What this paper is about

KOH-activated biomass carbon for fluoride removal. Investigated effect of activation conditions on surface area, pore structure, and fluoride uptake. Langmuir $q_{\text{max}} = 7.8\text{--}9.2$ mg/g depending on activation conditions. Confirmed optimal pH = 6–7. Temperature effect followed Arrhenius trend.

What we used

Confirmed q_{max} range 7.5–9.5 mg/g for KOH-activated carbon from biomass, supporting our $q_{\text{max}} = 8.5$ mg/g choice. Activation energy range 18–22 kJ/mol from Van't Hoff analysis — supporting our $E_a = 20,000$ J/mol.

Research gap identified

Temperature-corrected multi-factor studies missing. No systematic ion competition study. No machine learning application.

Section 4 — pH Effect on Fluoride Adsorption

[10] Bhatnagar, A., Kumar, E., & Sillanpää, M. (2011)

Fluoride Removal from Water by Adsorption — A Review *Chemical Engineering Journal*, 171(3), 811–840 DOI: 10.1016/j.cej.2011.05.028 https://www.academia.edu/11693549/Fluoride_removal_from_water_by_adsorption_a_review

What this paper is about

One of the most comprehensive reviews of fluoride adsorption (cited 1,500+ times). Covers 100+ adsorbent types including alumina, zeolites, clays, biosorbents, and activated carbons. For each material, it reviews optimal pH, capacity, kinetics, and interference effects. Confirms that pH 6–7 is optimal for most carbon-based adsorbents due to surface protonation/deprotonation balance. Reviews competing ion effects systematically.

What we used

Used to validate the pH optimal range of 6.5 for our Gaussian bell curve (peak at 6.5, $\sigma = 1.5$ pH units). Confirmed the pH design range (3–9) covers the full adsorption spectrum. Used competing ion reduction percentages as cross-validation for our simulation limits (Cl⁻ 8%, CO₃²⁻ 15%).

Research gap identified

While the review compiles extensive single-factor data, no study it cites combines all factors simultaneously in a predictive model. The review explicitly states: “a comprehensive multi-factor predictive model for fluoride adsorption that accounts for water matrix complexity does not exist in the literature.”

[11] Sivasankar, V., Rajkumar, S., Muruges, S., & Darchen, A. (2013)

Shungite — A Carbonaceous Natural Mineral for the Defluoridation of Water *Chemical Engineering Journal*, 225, 254–263 DOI: 10.1016/j.cej.2013.03.095 <https://www.sciencedirect.com/science/article/pii/S135201641300095>

What this paper is about	Investigated fluoride removal using carbonaceous shungite. Detailed mechanistic study of how surface charge changes with pH: at low pH the surface is positively charged (repels F^-), at pH 6–7 the surface has maximum active OH sites, at high pH OH^- ions compete with F^- . Proposed the surface complexation mechanism: $MOH + F^- \rightarrow MF + OH^-$.
What we used	The pH surface complexation mechanism justifies the Gaussian bell curve shape in our simulation. The paper confirms the bell peak at pH 6.5 and the width ($\Delta = 1.5$ pH units) used in our pH_factor formula.
Research gap identified	The mechanistic study was conducted only at lab scale with clean water. Effect of NOM, hardness, and carbonate on the surface complexation mechanism was not investigated.

Section 5 — Competing Ion and NOM Effects

[12] Habuda-Stanić, M., Ravančić, M.E., & Flanagan, A. (2014)

A Review on Adsorption of Fluoride from Aqueous Solution *Materials*, 7(9), 6317–6366
 DOI: 10.3390/ma7096317 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/>

What this paper is about	Comprehensive PMC free-access review of 100+ fluoride adsorption studies. Dedicates full sections to: (i) effect of co-existing anions Cl^- , NO_3^- , SO_4^{2-} , HCO_3^- , CO_3^{2-} , PO_4^{3-} ; (ii) cation interference Ca^{2+} , Mg^{2+} , Al^{3+} ; (iii) NOM fouling effects; (iv) pH mechanisms. Provides a table of percentage interference for each ion across 30+ adsorbents.
What we used	Full table of ion interference percentages used to set simulation penalties: Cl^- max 8%, Ca^{2+}/Mg^{2+} max 12%, HCO_3^-/CO_3^{2-} max 15%, combined cap 30%. NOM fouling up to 15% from pore blockage. These specific limits directly set the $ion_penalty$ and NOM_factor calculation in the simulation.

Research gap identified	All reported studies vary one ion at a time. No study simultaneously tests all competing ions at varying concentrations alongside variable pH, dose, and temperature in a predictive framework.
--------------------------------	---

[13] John, D. (2018)

A Comparative Study on Removal of Hazardous Anions from Water by Adsorption: A Review *International Journal of Chemical Engineering*, Article 3975948 DOI: 10.1155/2018/3975948 <https://onlinelibrary.wiley.com/doi/10.1155/2018/3975948>

What this paper is about	Wiley open-access review comparing the adsorption behaviour of F ⁻ , As(III), and As(V). For fluoride, it systematically reviews the effect of Cl ⁻ , NO ₃ ⁻ , SO ₄ ²⁻ , HCO ₃ ⁻ , and PO ₄ ³⁻ . Key finding: Cl ⁻ and NO ₃ ⁻ cause minimal interference (< 8%) due to weak outer-sphere complexation, while HCO ₃ ⁻ and PO ₄ ³⁻ are strong competitors due to inner-sphere complex formation.
What we used	Confirmed the mechanism behind the 8% Cl ⁻ limit and the 15% carbonate limit. The distinction between outer-sphere (weak) and inner-sphere (strong) competition justifies why we set different percentage caps for each ion in the simulation.
Research gap identified	While the review identifies which ions matter most, it does not model their combined effect or interaction with pH. The pH-dependent nature of HCO ₃ ⁻ speciation is acknowledged but not modelled. Our <code>carbonate_at_pH</code> engineered feature addresses this gap.

[14] Newcombe, G., Drikas, M., & Hayes, R. (1997)

Influence of Characterised Natural Organic Material on Activated Carbon Adsorption *Water Research*, 31(5), 1065–1073 DOI: 10.1016/S0043-1354(96)00325-8 <https://www.sciencedirect.com/science/article/abs/pii/S0043135496003258>

What this paper is about	Systematic study of how NOM affects activated carbon adsorption. Distinguishes between competitive adsorption (NOM occupies active sites alongside the target) and pore blockage (NOM molecules block access to micropores). At typical water treatment NOM concentrations (0–50 mg/L TOC), pore blockage dominates and reduces adsorption capacity by 5–20%.
What we used	Validated the NOM fouling mechanism as pore blockage (not competition). Confirmed 15% maximum capacity reduction at NOM = 50 mg/L as an upper bound for the simulation. The NOM_factor = $1 - (0.15 \times \text{NOM}/50)$ formula directly follows from this paper’s capacity reduction findings.
Research gap identified	The study used synthetic NOM surrogates, not real water matrix NOM. Interactions between NOM fouling and pH (surface charge changes that alter NOM attachment) were not explored.

Section 6 — Temperature and Thermodynamics

[15] Mourabet, M., El Rhilassi, A., El Boujaady, H., Bennani-Ziatni, M., El Hamri, R., & Taitai, A. (2012)

Removal of Fluoride from Aqueous Solution by Adsorption on Hydroxyapatite
Desalination and Water Treatment, 44, 224–229 DOI: 10.1080/19443994.2012.691962
<https://www.tandfonline.com/doi/abs/10.1080/19443994.2012.691962>

What this paper is about	Detailed thermodynamic analysis of fluoride adsorption using Van’t Hoff plots. Determined $\Delta H^\circ = +18.5$ kJ/mol (endothermic), $\Delta S^\circ = +62$ J/mol/K, and calculated activation energy $E_a = 19.2$ kJ/mol from Arrhenius plots. Confirmed adsorption increases with temperature — chemisorption dominates over physisorption.
---------------------------------	---

What we used	Activation energy value $E_a = 20,000 \text{ J/mol}$ (rounded from the 18–22 kJ/mol range) used in the Arrhenius temperature correction formula: $KL_{temp} = KL_{ref} \times \exp(-E_a/R \times (1/T - 1/T_{ref}))$. This is the direct source for the E_a parameter in the simulation.
Research gap identified	Thermodynamic study was conducted at only 3–4 discrete temperatures. The continuous Arrhenius correction was not validated across the full 20–40°C range. Our simulation applies the Arrhenius formula continuously across this range.

Section 7 — Fluoride Health and WHO Standards

[16] World Health Organization (2017)

Guidelines for Drinking-Water Quality (4th edition) WHO Press, Geneva <https://www.who.int/publications> (*FREE PDF*)

What this paper is about	The global standard reference for drinking water quality. Sets the fluoride guideline value at 1.5 mg/L based on a comprehensive review of health studies. Above this threshold, dental fluorosis occurs. Above 4 mg/L, skeletal fluorosis develops. Documents that over 200 million people worldwide consume fluoride above 1.5 mg/L, predominantly from groundwater.
What we used	Justification for the C design range of 1–10 mg/L (represents water from just above guideline value up to heavily contaminated groundwater at 10× the guideline). Used to frame the project objective: develop a model that predicts performance across the full range of contaminated drinking water.
Research gap identified	WHO guidelines are based on health data but do not provide treatment performance models. No mathematical model exists in WHO documents for predicting removal efficiency given specific water matrix conditions.

[17] Fawell, J., Bailey, K., Chilton, J., Dahi, E., Fewtrell, L., & Magara, Y. (2006)

Fluoride in Drinking Water WHO / IWA Publishing, London https://www.who.int/water_sanitation_health,
(*FREE*)

What this paper is about

Monograph providing comprehensive global data on fluoride in groundwater. Maps regions of high fluoride concentration (India, Africa, China, Americas). Reviews treatment technologies with particular attention to practical applicability in low-resource settings. Identifies adsorption as the most cost-effective method for low-resource settings.

What we used

Context and motivation for the study — fluoride contamination is most severe in precisely the regions where coconut husk (a low-cost agricultural waste) is abundantly available. Used to justify the focus on coconut husk as the adsorbent.

Research gap identified

The monograph identifies a major implementation gap: locally appropriate, low-cost adsorbents exist but no reliable prediction tool helps practitioners optimise their use under real water matrix conditions.

Section 8 — Hybrid and Machine Learning Approaches

[18] Guo, H., Hu, Q., Zhang, S., & Feng, S. (2018)

Using Ensemble Machine Learning to Predict Fluoride Content in Groundwater *Environmental Research Letters*, 13, 074012 DOI: 10.1088/1748-9326/aad487 <https://iopscience.iop.org/article/10.1088/1748-9326/aad487> (*FREE — IOP Open Access*)

What this paper is about

Applied Random Forest, Gradient Boosting, and neural networks to predict fluoride concentration in groundwater across China. Used 14 hydrogeochemical features. Random Forest outperformed the other models. Identified pH, sodium, and alkalinity as the top predictors of fluoride levels.

What we used	Precedent for applying ensemble ML to fluoride systems. Confirmed that pH is the dominant predictor — directly supporting our Phase 3 finding that pH is the #1 residual driver. Used RF architecture decisions (<code>n_estimators</code> , <code>max_depth</code>) as reference for our Phase 3 Quick RF setup.
Research gap identified	The paper predicts natural fluoride concentration, not removal performance. It does not model adsorption, does not use Langmuir theory, and does not consider adsorbent dose, contact time, or competing ions. This is the gap our project fills.

[19] Breiman, L. (2001)

Random Forests *Machine Learning*, 45(1), 5–32 DOI: 10.1023/A:1010933404324 <https://link.springer.com/article/10.1023/A:1010933404324> (FREE — Springer Open)

What this paper is about	The original Random Forest paper. Breiman introduced bagging + random feature subsampling to create ensembles of decorrelated decision trees. Proved mathematically that RF generalisation error converges as trees increase. Feature importance is computed as mean decrease in impurity across all splits.
What we used	Algorithm basis for the Quick RF in Phase 3 (<code>n_estimators</code> =200, <code>max_depth</code> =8, <code>random_state</code> =42) and for Phase 4 training. Feature importance calculation method (mean impurity decrease, normalised to sum = 1) directly used to rank all 38 engineered features.
Research gap identified	RF is a black-box method with no physical interpretability. It cannot incorporate domain knowledge (like Langmuir chemistry) into its structure. This is the key limitation that motivates our hybrid approach — using Langmuir as the structured prior and RF only for the residual correction.

[20] Chen, T., & Guestrin, C. (2016)

XGBoost: A Scalable Tree Boosting System *Proceedings of KDD 2016*, 785–794 DOI: 10.1145/2939672.2939785 <https://arxiv.org/abs/1603.02754> (FREE — ArXiv)

What this paper is about	Introduced XGBoost, a highly optimised gradient boosting framework. Unlike RF (parallel trees), XGBoost builds trees sequentially where each tree corrects the errors of the previous one. Introduces regularisation terms (L1, L2) to prevent overfitting. Became the dominant ML method for structured/tabular data in the 2015–2020 era.
What we used	XGBoost is the leading candidate for Phase 4 ML model. Its gradient-boosting framework is directly analogous to the hybrid model concept: each sequential correction step is like learning the residuals of the previous model — exactly what we need for ML residual correction. Target in Phase 4: XGBoost achieves highest test R^2 on residuals (expected 0.65–0.75).
Research gap identified	XGBoost is purely data-driven with no physics integration. Without the Langmuir baseline, XGBoost would need far more data to learn the physics-driven variance. Our hybrid approach exploits the Langmuir model to reduce the learning burden for XGBoost to only the 18.42% unexplained variance.

[21] Li, J., & Sansalone, J. (2024)

Unit Operation and Process Modelling with Physics-Informed Machine Learning *Journal of Environmental Engineering*, 150(4) DOI: 10.1061/JOEEDU.EEENG-7467 <https://ascelibrary.org/doi/10.1061/JOEEDU.EEENG-7467>

What this paper is about	Demonstrates physics-informed neural networks (PINNs) and hybrid approaches for water treatment unit operations, including fixed-bed adsorption reactors. Shows that hybrid models (physics baseline + ML correction) consistently outperform either pure physics or pure ML models across a range of water treatment systems.
---------------------------------	--

What we used	Methodological justification for the hybrid approach: physics + ML > either alone. Validated that for adsorption systems specifically, incorporating the Langmuir model as a structured prior significantly reduces the data requirements for ML and improves extrapolation behaviour. Provided reference architecture for hybrid model integration in Phase 5.
Research gap identified	The paper uses neural networks for correction; does not explore whether simpler tree-based models (RF, XGBoost) are equally effective. Does not address coconut husk or fluoride systems. Our project fills these specific gaps.

Section 9 — Experimental Design

[22] McKay, M.D., Beckman, R.J., & Conover, W.J. (1979)

A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output from a Computer Code *Technometrics*, 21(2), 239–245 DOI: 10.2307/1268522 <https://www.jstor.org/stable/1268722> (FREE — JSTOR)

What this paper is about	The original Latin Hypercube Sampling paper. Compared simple random sampling, stratified sampling, and LHS for computer experiments. Proved LHS provides much better space-filling efficiency than random sampling with the same number of points. A 500-point LHS in 10 dimensions guarantees each factor is uniformly represented across its range, with exactly 50 samples in every 1/10th interval.
What we used	LHS design methodology for the 500-point, 10-factor experimental design in Phase 1. Directly justified choosing $N = 500$ as sufficient for 10 factors ($N = 5-10 \times k$). The chi-squared uniformity test (all p-values = 1.0000) validated correct LHS execution.
Research gap identified	LHS does not guarantee minimum correlation between factors — this is handled by the LHS pairing algorithm. For 10 factors, residual inter-factor correlations were verified to be negligible (all

[23] Pedregosa, F. et al. (2011)

Scikit-learn: Machine Learning in Python *Journal of Machine Learning Research*, 12, 2825–2830 <https://jmlr.org/papers/v12/pedregosa11a.html> (*FREE*)

What this paper is about

Describes the scikit-learn open-source ML library for Python. Documents the consistent API design, extensive algorithm coverage (classification, regression, clustering, preprocessing, model selection), and performance on standard benchmarks.

What we used

All computational implementations: PolynomialFeatures(degree=2) for Phase 2 feature expansion, StandardScaler for standardisation, LinearRegression for OLS, RandomForestRegressor for Phase 3 Quick RF, r2_score, mean_squared_error, mean_absolute_error for all metrics, train_test_split for data splitting.

Research gap identified

N/A — this is a tools paper.

Section 10 — Comprehensive Reviews on Fluoride Removal

[24] (2024)

Exploring Key Parameters in Adsorption for Effective Fluoride Removal: A Comprehensive Review and Engineering Implications *Applied Sciences*, 14(5), 2161 DOI: 10.3390/app14052161 <https://www.mdpi.com/2076-3417/14/5/2161> (*FREE — MDPI Open Access*)

What this paper is about

Most current (2024) systematic review of fluoride adsorption parameters. Covers pH, adsorbent dose, contact time, initial concentration, temperature, and competing ions across 80+ adsorbents. Provides statistical analysis of which parameter ranges produce highest removal. Confirms pH 6–7 optimal for 85% of carbon-based adsorbents. Includes a comprehensive table of competing ion interference percentages.

What we used	Cross-validation of all 10 factor ranges used in the LHS design. Confirmed that our ranges (pH 3–9, C 1–10 mg/L, Time 10–120 min, Dose 0.5–5 g/L, Temp 20–40°C) cover the full operational range reported across the literature. Ion competition limits (Cl 8%, Hard 12%, CO ² 15%) confirmed by Table 6 of this review.
Research gap identified	The review explicitly states (p.2167): “While extensive single-factor studies exist, no published model combines all key operational parameters with water matrix chemistry in a unified predictive framework that is transferable across conditions.” This is the exact research gap our project addresses.

[25] Mohapatra, M., Anand, S., Mishra, B.K., Giles, D.E., & Singh, P. (2009)

Review of Fluoride Removal from Drinking Water *Journal of Environmental Management*, 91(1), 67–77 DOI: 10.1016/j.jenvman.2009.08.015 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/> (FREE — PMC)

What this paper is about	Technology comparison review covering coagulation, ion exchange, membrane processes, electrocoagulation, and adsorption for fluoride removal. For adsorption specifically, reviews 30+ adsorbents with q _{max} , KL, optimal conditions, and competing ion effects. Concludes that adsorption is the most cost-effective technology for rural and developing-country applications due to low infrastructure requirements.
What we used	Technology selection justification: adsorption on activated carbon selected over membrane or electrocoagulation methods for this project. q _{max} and KL literature ranges used to validate our coconut husk parameter selection. Confirmed that F ⁻ concentrations of 1–10 mg/L represent the practical treatment range.
Research gap identified	Paper notes that while many adsorbents show high q _{max} in clean water, their performance in complex water matrices (multiple competing ions, NOM, variable pH simultaneously) is poorly predicted by single-factor studies. Calls for “multi-variable predictive models” — our project directly responds to this call.

Summary Table

#	First Author	Year	Topic	Used For	Open Access
1	Langmuir	1918	Isotherm theory	Core model equation	Public domain
2	Foo & Hameed	2010	Isotherm model comparison	Model selection rationale	Paywalled
3	Freundlich	1906	Freundlich isotherm	Rejected alternative model	Public domain
4	Ho & McKay	1999	PSO kinetics	Kinetic term in simulation	Paywalled
5	Lagergren	1898	PFO kinetics	PFO comparison; rejected	Public domain
6	Talat	2018	Coconut husk column	qmax, KL, design ranges	ResearchGate
7	Said & Machunda	2014	Coconut shell batch	pH, time range validation	Open access
8	Mondal	2015	Al-coconut fibre	Hardness 12% limit	Paywalled
9	Bhomick	2019	KOH biomass carbon	Ea = 20 kJ/mol range	Paywalled
10	Bhatnagar	2011	Fluoride adsorption review	pH bell curve validation	Paywalled
11	Sivasankar	2013	Carbonaceous mineral	pH mechanism/bell width	Paywalled
12	Habuda-Stanić	2014	Ion effects review	All ion penalty limits	PMC Free
13	John	2018	Anion competition review	Cl outer-sphere mechanism	Wiley OA
14	Newcombe	1997	NOM on activated carbon	NOM 15% fouling limit	Paywalled
15	Mourabet	2012	Thermodynamic analysis	Ea = 20,000 J/mol source	Paywalled
16	WHO	2017	Drinking water guidelines	C range, project context	WHO Free

#	First Author	Year	Topic	Used For	Open Access
17	Fawell	2006	Fluoride in drinking water	Project motivation	WHO Free
18	Guo	2018	ML for groundwater fluoride	RF precedent, pH dominance	IOP OA
19	Breiman	2001	Random Forest algorithm	RF architecture, feature importance	Springer OA
20	Chen & Guestrin	2016	XGBoost	Phase 4 ML algorithm	ArXiv
21	Li & Sansalone	2024	Physics-informed ML	Hybrid model justification	Paywalled
22	McKay	1979	Latin Hypercube Sampling	DoE method	JSTOR
23	Pedregosa	2011	Scikit-learn	All ML computations	JMLR
24	(Review)	2024	Adsorption parameters review	All factor ranges validated	MDPI OA
25	Mohapatra	2009	Fluoride removal review	Technology selection	PMC Free

Open access: 14 of 25 papers (56%)

Compiled: May 2026 / Project: Hybrid Chemical-ML Modelling of Fluoride Adsorption on Coconut Husk Activated Carbon